MATH2012
Fundamental Concepts of Mathematics

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Chapter Outline

Chapter 1. Set Theory

Chapter 2. Logic

Chapter 3. Techniques of Proof

Chapter 4. Relations

Chapter 5. Functions

Chapter 6. Introduction to Calculus: Sequences and Series (optional)

Chapter 7. Introduction to Group Theory (optional)

Reference Books

Main Reference

1. Class lecture notes — This course will follow closely with the lecture notes which can be downloaded from moodle.

2. G. Chartrand, A.D. Polimeni and P. Zhang, *Mathematical Proofs: A Transition to Advanced Mathematics*, Pearson/Addison-Wesley

3. R. Courant and H. Robbins, *What is Mathematics? An Elementary Approach to Ideas and Methods*, Oxford University Press, 2nd edition, revised by I. Stewart

Suggested Readings

1. P.R. Halmos, *Naive Set Theory*, Springer-Verlag

2. K. Houston, *How to Think Like a Mathematician: A Companion to Undergraduate Mathematics*, Cambridge University Press

3. K.T. Leung and D.L.C. Chen, *Elementary Set Theory*, Hong Kong University Press.

4. K.T. Leung, P.H. Cheung, *Fundamental Concepts of Mathematics*, Hong Kong University Press.

5. C. Schumacher, *Chapter Zero: Fundamental Notions of Abstract Mathematics*, 2nd Edition, Addison-Wesley

6. I. Stewart, *Concepts of Modern Mathematics*, Dover Publications

Chapter 1. Set Theory

1.1. Sets

One of the most fundamental concepts in modern mathematics is the theory of sets. The theory of sets is usually considered to have begun formally with a German mathematician Georg Cantor (1845-1918) during the latter part of the nineteenth century. In this chapter we shall adopt an informal attitude and introduce set theory as a mathematical language which we use for reading, writing and communication.

The basic concept in set theory is a *set* which we conceive as a collection of *distinct* objects. An *element* is one of the objects that lies in the set. We say that the element *belongs* to the set. Sets are conventionally denoted by capital letters, whereas their elements are denoted by small letters. We write

$$x \in S$$

to mean x belongs to the set S. If the object x does not belong to the set S, we write $x \notin S$. When two objects x and y are *equal*, denoted by $x = y$, we mean that x and y are just different names for the same object.

To describe a set, we may either use any of the following methods:

- **Word description**

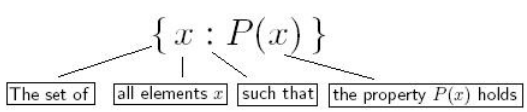
  For example, let S be the set of colors of a rainbow.

- **Listing**

  Elements of a set are listed within a pair of braces. For example,

  $$S := \{ \text{ Red, Orange, Yellow, Green, Blue, Indigo, Violet } \}.$$

- **Set-builder notation**[1]

  For example,

  $$S := \{x : x \text{ is the color of a rainbow}\}$$

A set contains only *distinct* elements. Thus, $\{2, 2, 3\} = \{2, 3\}$. Moreover, the order in which the elements are listed does not matter. Thus $\{2, 3\} = \{3, 2\}$. Two sets are *equal* if they contain exactly the same elements.[2] Thus,

$$\{2, 4, 6, 8, 10\} = \{x : \ 0 < x \leq 10 \text{ and } x \text{ is an even integer.}\}$$

[1] Such notation is related to the *axiom of specification*, which is one of the axioms of *Zermelo-Fraenkel axiomatic set theory*.

[2] This is known as the *axiom of extension* in the Zermelo-Fraenkel axiomatic set theory.

In other words, two sets A and B are equal if every element of A is also an element of B and every element of B is also an element of A.

Example 1.1 Show that the sets $A = \{x : x^2 - 4 = 0\}$ and $B = \{x : |x| = 2\}$ are equal.

Solution. Let us check if A and B contain exactly the same elements by representing them by the listing method. Solving the equation $x^2 - 4 = 0$ yields $x = 2$ or -2, so we have $A = \{2, -2\}$. Likewise, solving the equation $|x| = 2$ also yields $x = 2$ or -2, so we have $B = \{2, -2\}$ again and hence $A = B$. $\square$

Exercise Suppose x and y are different objects. Determine if each of the following statements is true or false:

(a) $x \in \{x\}$

(b) $x \in \{\{x\}\}$

(c) $\{x\} \in \{\{x\}\}$

(d) $\{x\} = \{x, x\}$

(e) $x \in \{\{x\}, y\}$

(f) $\{x, x\} \in \{\{x\}, y\}$

The *empty set*, denoted by $\emptyset$ (or $\{\}$), is the set which contains no element. For example,

$$\{x : x \text{ is a positive integer and } x + 1 = 0\} = \emptyset.$$

Note that $\{\emptyset\}$ is not the empty set. It is a set containing one element, namely $\emptyset$.

If a set has a finite number of elements, then it is called a *finite set*, otherwise it is called an *infinite set*. If S is a finite set, we use $|S|$, the *cardinality of S*, to denote the number of elements contained in S.

Exercise Find the cardinality of each of the following sets:

$$A = \{1, 2\}, \quad B = \{\{1, 2\}\}, \quad C = \{1, 2, \{1, 2\}, \emptyset\}, \quad D = \emptyset$$

If S is an infinite set, then it becomes difficult to define the cardinality of S. We shall come back to this later in this course. For the moment we could just say that the cardinality is undefined for infinite sets.

The most familiar examples of infinite sets are given by sets of numbers. We shall denote the set of all natural numbers, integers, rational numbers, irrational numbers, real numbers and complex numbers by $\mathbb{N}$[3], $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{I}$[4], $\mathbb{R}$ and $\mathbb{C}$ respectively.

[3] We shall not treat 0 as a natural number in this course.

[4] There is no common symbol to denote the set of irrational numbers, but we shall use this in our course

1.2. Subsets

We now introduce a relationship between sets.

Definition 1.2 Let A and B be sets. If every element of A is an element of B, then A is a *subset* of B, written as $A \subseteq B$. If A is NOT a subset of B, then we write $A \nsubseteq B$.

Remark 1.3 Two sets A and B are equal "is equivalent to" (or "if and only if") $A \subseteq B$ and $B \subseteq A$.

Example 1.4 $\{1\} \subseteq \{1, 2, a, b\}$ and $\{1, 2\} \subseteq \{1, 2, a, b\}$, but $\{1, 2, 3\} \nsubseteq \{1, 2, a, b\}$.

Example 1.5

(i) $\{-7, 4, 11\} \nsubseteq \{x : x \text{ is an odd integer}\}$.

(ii) The set of all even integers is a subset of $\mathbb{Z}$.

(iii) $\left\{3, \frac{1}{7}, \frac{29}{9}\right\} \subseteq \mathbb{Q}$.

(iv) For any set S, we have $S \subseteq S$.

(v) For any set S, we have $\emptyset \subseteq S$.

Definition 1.6 We say that A is a *proper subset* of B if $A \subseteq B$ and $A \neq B$. In this case, we write $A \subset B$.

Example 1.7

(i) $\{1, 2\} \subset \{1, 2, a, b\}$.

(ii) For any set S, the subset S is NOT a proper subset of S.

(iii) For any nonempty set S, we have $\emptyset \subset S$.

(iv) For sets of numbers, we have $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Exercise Determine if each of the following statements is true or false:

(a) $\{0\} \subseteq \{\{0\}, \{1\}\}$

(b) $\{0, 1\} \subseteq \{\{0, 1\}\}$

(c) $\emptyset \subseteq \{\emptyset\}$

(d) $\emptyset \subset \{\emptyset\}$

(e) $\{\emptyset\} \subseteq \{\emptyset\}$

(f) $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$

Interval notation

Intervals are frequently encountered subsets of $\mathbb{R}$. If a and b are real numbers with $a < b$, then

1. $(a, \infty) := \{x \in \mathbb{R} : a < x\}$.
2. $(-\infty, a) := \{x \in \mathbb{R} : x < a\}$.
3. $[a, \infty) := \{x \in \mathbb{R} : a \leq x\}$.
4. $(a, b) := \{x \in \mathbb{R} : a < x < b\}$.

Similarly, we can define $(-\infty, a]$, $[a, b)$, $(a, b]$ and $[a, b]$. These sets are called *intervals*. Intervals of the type (a, b) are called *open intervals* and the endpoints are excluded in the set. Intervals of the type $[a, b]$ are called *closed intervals* and the endpoints are included in the set. Intervals using both square and round brackets are called *half-closed intervals* or *half-open intervals*.

Example 1.8 The set

$$\{x \in \mathbb{R} : -1 < x \leq 4\}$$

can be written as $(-1, 4]$. □

Example 1.9 The open interval $(0, 1)$ is a subset of $[0, 1)$, so $(0, 1) \subseteq [0, 1)$. We also have $(0, 1) \subseteq (0, 1]$, $(0, 1) \subseteq [0, 1]$. But $(0, 1] \not\subseteq [0, 1)$. □

Example 1.10 Find the sets of (real) solutions, or simply *solution sets*, of (i) $2x + 3 < 0$, (ii) $x^2 + 2x - 3 \leq 0$, (iii) $x^2 + 1 \geq 0$ and (iv) $x^2 + 1 < 0$.

Solution. (i) $(-\infty, -3/2)$, (ii) $[-3, 1]$, (iii) $\mathbb{R}$ and (iv) $\emptyset$. □

1.3. Universal set and Venn diagram

Definition 1.11 A *universal set*, denoted by U, is a set that contains all the elements being considered in a certain investigation.

The universal set can change from problem to problem. Any set can be a universal set, as long as you are studying that particular set. For example, if you are studying real numbers, then the set $\mathbb{R}$ will be the universal set and you can treat $\mathbb{R}$ as the "universe" and study everything inside.

Definition 1.12 Let A be a subset of the universal set U. Then the *complement* of A, denoted by A^c, is the set

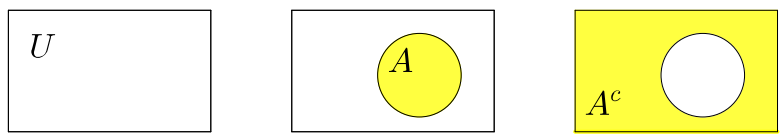
$$A^c := \{x \in U : x \notin A\}.$$

Example 1.13 Given that $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$. Then

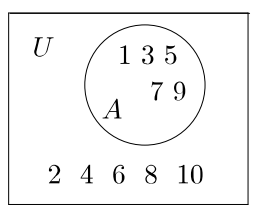
$$A^c = \{2, 4, 6, 8, 10\}.$$

The *Venn diagram* is a visual tool to investigate relations among subsets of a universal set.

- The universal set U is represented by the region enclosed by a rectangle (or other shape you like).
- Each subset of U is represented by a region inside the rectangle enclosed by a circle (or other shape you like).
- The region inside the rectangle but outside the circle is then representing A^c.



Example 1.14 We may represent the sets $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$ by a Venn diagram.



1.4. Set Operations

We introduce here several ways of creating new sets from old sets:

(i) Union of sets

Definition 1.15 Let A and B be sets. Then the *union* of A and B, denoted by $A \cup B$, is the set

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

Example 1.16 $\{1, x, y\} \cup \{x, z\} = \{1, x, y, z\}$ and $\{1, x, y\} \cup \emptyset = \{1, x, y\}$. □

Example 1.17 $\{x \in \mathbb{R} : x^2 + 2x - 3 > 0\} = (-\infty, -3) \cup (1, \infty)$. □

Example 1.18 In fact, the set of real numbers is just the union of the sets of rational and irrational numbers, that is, we have $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$. In particular, a real number is either a rational number or an irrational number. □

Theorem 1.19 *Let A, B and C be sets. Then the following statements hold:*

(a) $A \cup A = A$

(b) $A \cup B = B \cup A$ *(commutative law)*

(c) $A \cup (B \cup C) = (A \cup B) \cup C$ *(associative law)*

(d) $A \subseteq A \cup B$ *and* $B \subseteq A \cup B$

(e) $A \cup \emptyset = A$

Remark We shall study the formal proof and logic for this theorem (and some others) later in this course.

(ii) Intersection of sets

Definition 1.20 Let A and B be sets. Then the *intersection* of A and B, denoted by $A \cap B$, is the set

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

Example 1.21 $\{1, x, y\} \cap \{x, z\} = \{x\}$ and $\{1, y\} \cap \{x, z\} = \emptyset$. □

Two sets A and B are said to be *disjoint* if $A \cap B = \emptyset$, that is, the two sets have no element in common. The set of all rational numbers and the set of all irrational numbers are disjoint as no number can be both rational and irrational, so we have $\mathbb{Q} \cap \mathbb{I} = \emptyset$.

Theorem 1.22 *Let A, B and C be sets. Then the following statements hold:*

(a) $A \cap A = A$

(b) $A \cap B = B \cap A$ *(commutative law)*

(c) $A \cap (B \cap C) = (A \cap B) \cap C$ *(associative law)*

(d) $A \cap B \subseteq A$ *and* $A \cap B \subseteq B$

(e) $A \cap \emptyset = \emptyset$

Theorem 1.23 (Distributive Laws) *Let A, B and C be sets. Then the following statements hold:*

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(iii) Complement of A in B

Given any two sets, we can consider a set whose elements belong to one of these two sets but not to the other.

Definition 1.24 Let A and B be sets. Then the *complement* of A in B, denoted by $B \setminus A$, is the set

$$\{x : x \in B \text{ and } x \notin A\}.$$

Remark If U is the universal set and A is a subset of U, then we have $A^c = U \setminus A$.

Example 1.25 $\{1, x, y\} \setminus \{x, z\} = \{1, y\}$ and $\{1, y\} \setminus \{x, z\} = \{1, y\}$. □

Example 1.26 If $U = \mathbb{R}$ is the universal set, then $\mathbb{R} \setminus \mathbb{Q} = \mathbb{Q}^c = \mathbb{I}$. If $U = \mathbb{Z}$ is the universal set, then $\mathbb{Z} \setminus \mathbb{N} = \mathbb{N}^c = \{0, -1, -2, \dots\}$. □

Exercise Find

(a) $[1, 2] \setminus \{1\}$,

(b) $[1, 2] \setminus \{1, 2\}$,

(c) $[1, 2] \setminus \{1, 2, 3\}$,

(d) $[1, 2] \setminus (1, 1.5)$,

(e) $[1, 2] \setminus \emptyset$,

(f) $\mathbb{R} \setminus [1, 2]$.

Theorem 1.27 *Let A and B be sets. Then the following statements hold:*

(a) $A \setminus A = \emptyset$

(b) $A \setminus \emptyset = A$

(c) $\emptyset \setminus A = \emptyset$

(d) $A \setminus B = \emptyset$ *if and only if* $A \subseteq B$

(e) $(A \setminus B) \cap (B \setminus A) = \emptyset$

(f) $A \cap (B \setminus A) = \emptyset$

Theorem 1.28 (De Morgan's Laws) *Let A and B be subsets of a universal set U. Then*

$$(A \cup B)^c = A^c \cap B^c \quad \textit{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Equivalently, we have

$$U \setminus (A \cup B) = (U \setminus A) \cap (U \setminus B) \quad \textit{and} \quad U \setminus (A \cap B) = (U \setminus A) \cup (U \setminus B).$$

Example 1.29 Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$ and $B = \{1, 2, 3, 4, 5\}$. Then

$$A \cup B = \{1, 2, 3, 4, 5, 7, 9\}, \text{ and so } (A \cup B)^c = \{6, 8, 10\}.$$

Also,

$$A^c = \{2, 4, 6, 8, 10\}\,,\ B^c = \{6, 7, 8, 9, 10\}, \text{ and so } A^c \cap B^c = \{6, 8, 10\}.$$ □

(iv) Symmetric difference of A and B

Definition 1.30 Let A and B be sets. Then the *symmetric difference* of A and B, denoted by $A \triangle B$, is the set

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Example 1.31 $\{1, x, y\} \triangle \{x, z\} = \{1, y, z\}$ and $\{1, y\} \triangle \{x, z\} = \{1, x, y, z\}$. □

Theorem 1.32 *Let A and B be sets. Then the following statements hold:*

(a) $A \triangle A = \emptyset$

(b) $A \triangle \emptyset = A$

(c) $A \triangle B = (A \cup B) \setminus (A \cap B)$

(d) $A \triangle B = B \triangle A$

(v) Power sets

Definition 1.33 Let A be a set. The *power set* $\mathcal{P}(A)$ of A is the set of all subsets of A, that is,

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Example 1.34

(a) If $A = \{a\}$, then $\mathcal{P}(A) = \{\emptyset, \{a\}\}$.

(b) If $A = \{a, b\}$, then $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

(c) If $A = \emptyset$, then $\mathcal{P}(A) = \{\emptyset\}$. □

Exercise Find the power set of each of the following sets:

(a) $A = \{a, b, c\}$

(b) $A = \{a, \{a\}\}$

(c) $A = \mathcal{P}(\{a\})$

Theorem 1.35 *Let A be a finite set with $|A| = n$. Then $|\mathcal{P}(A)| = 2^n$.*

(vi) Cartesian products of sets

In analytic geometry, we are familiar with the use of *ordered pairs* of numbers, say $(1, 2)$, to specify points on the plane. The crucial feature of ordered pairs is that the order in which the numbers is listed is important, and that for two ordered pairs (x, y) and (w, z) to be equal, that is, $(x, y) = (w, z)$, we must have $x = w$ and $y = z$. So if $x \neq y$, then $(x, y) \neq (y, x)$. Using the concept of ordered pairs, we can introduce the concept of *Cartesian product* of two sets:

Definition 1.36 Let A and B be two sets. The *Cartesian product* of A and B is the set

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

Example 1.37 Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$. Then

$$A \times B = \{(1, x),\ (1, y),\ (2, x),\ (2, y),\ (3, x),\ (3, y)\}$$

and

$$B \times A = \{(x, 1),\ (x, 2),\ (x, 3),\ (y, 1),\ (y, 2),\ (y, 3)\}.$$ □

Note that in general $A \times B \neq B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$. In the last case we may also write A^2 to denote $A \times A$. If $A = B = \mathbb{R}$, then

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R},\ y \in \mathbb{R}\}$$

is the set of ordered pairs (x, y) of real numbers, and is usually represented by the coordinate plane with x and y being the coordinates of a point with respect to a pair of axes.

In the same way, one may define the cartesian product of three sets, or more generally, the cartesian product of k sets:

$$A_1 \times A_2 \times \cdots \times A_k = \{(a_1, a_2, \ldots, a_k) : a_1 \in A_1,\ a_2 \in A_2, \ldots, a_k \in A_k\}.$$

1.5. An Application: Counting Principle

Recall that if S is a finite set, the cardinality $|S|$ of S is the number of elements contained in S.

Theorem 1.38 *For any finite sets A, B and C, we have*

$$|A \cup B| = |A| + |B| - |A \cap B|$$

and

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Example 1.39 Let $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$ and $C = \{2, 3, 7, 8\}$. Then

	Set	Cardinality
A	$\{1, 2, 3, 4, 5\}$	5
B	$\{2, 4, 6, 8\}$	4
C	$\{2, 3, 7, 8\}$	4
$A \cap B$	$\{2, 4\}$	2
$A \cap C$	$\{2, 3\}$	2
$B \cap C$	$\{2, 8\}$	2
$A \cap B \cap C$	$\{2\}$	1

By Theorem 1.38, $|A \cup B \cup C| = 5 + 4 + 4 - 2 - 2 - 2 + 1 = 8$, corresponding to the eight elements of $A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$. □

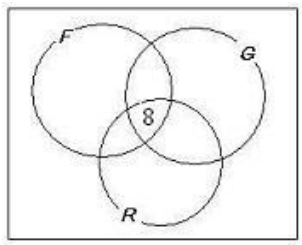
Example 1.40 Suppose that 100 of the 120 mathematics students take at least one of the languages French, German and Russian. Given that

- 65 take French, 45 take German, 42 take Russian
- 20 take both French and German
- 25 take both French and Russian
- 15 take both German and Russian.

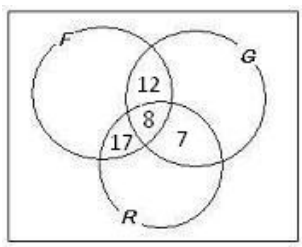
Let F, G, R be the set of mathematics students who take French, German and Russian respectively. From the data above, we have

$$|F \cup G \cup R| = 100,\ |F| = 65,\ |G| = 45,\ |R| = 42,\ |F \cap G| = 20,$$
$$|F \cap R| = 25,\ |G \cap R| = 15.$$

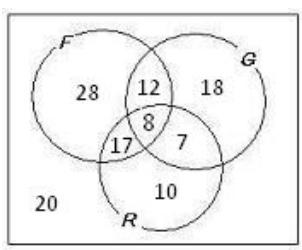
By Theorem 1.38, the number of mathematics students who take all three languages is therefore $|F \cap G \cap R| = 100 - 65 - 45 - 42 + 20 + 15 + 25 = 8$.



It follows that $|(F \cap G) \backslash R| = 20 - 8 = 12$, $|(F \cap R) \backslash G| = 25 - 8 = 17$ and $|(G \cap R) \backslash F| = 15 - 8 = 7$.



As a result, $|F \backslash (G \cup R)| = 28$, $|G \backslash (F \cup R)| = 18$ and $|R \backslash (F \cup G)| = 10$. Finally, $|\,(F \cup G \cup R)^c\,| = 120 - 100 = 20$.



Theorem 1.41 (Inclusion-Exclusion Principle) *For any finite sets $A_1, A_2, \ldots A_n$, we have*

$$\begin{aligned} |A_1 \cup A_2 \cup \cdots \cup A_n| &= \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| \\ &\quad - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|. \end{aligned}$$

The Inclusion-Exclusion Principle is a generalization of Theorem 1.38. It has many applications in combinatorics.